Experiment 6.1

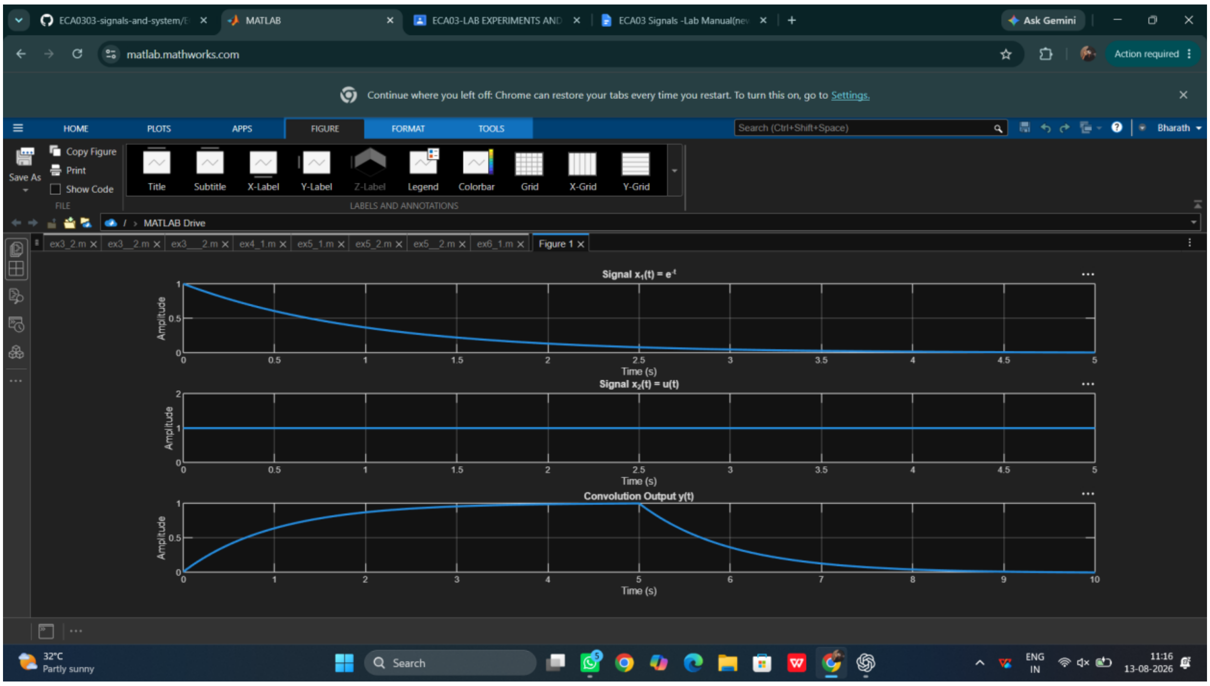
Convolution of Continuous Time Signals

Program:

```
% Convolution of Continuous-Time Signals

clc;

clear;

close all;

% Time axis

t = 0:0.01:5;

% Input signals

x1 = exp(-t); % Exponential signal x2

= ones(size(t)); % Unit step signal

% Convolution

y = conv(x1,x2)*0.01;

% Time axis for output

t_conv = 0:0.01:(length(y)-1)*0.01;

% Plotting

figure;

subplot(3,1,1);

plot(t,x1,'LineWidth',2);

grid on;

title('Signal x_1(t)=e^{-t}');

xlabel('Time (s)');

ylabel('Amplitude');

subplot(3,1,2);

plot(t,x2,'LineWidth',2);
```

```
grid on;
title('Signal x_2(t)=u(t)');
xlabel('Time (s)');
ylabel('Amplitude');
subplot(3,1,3);
plot(t_conv,y,'LineWidth',2);
grid on;
title('Convolution Output y(t)');
xlabel('Time (s)');
ylabel('Amplitude');
```

Output Waveform:



Result:

The convolution of the continuous-time signals $x1(t)=e-tu(t)$ and $x2(t)=u(t)$ was performed successfully using MATLAB. The output signal $y(t)=1-e-t$ was obtained and plotted, demonstrating the convolution operation for continuous-time systems.